EXPERIMENT NO: 1.(a)

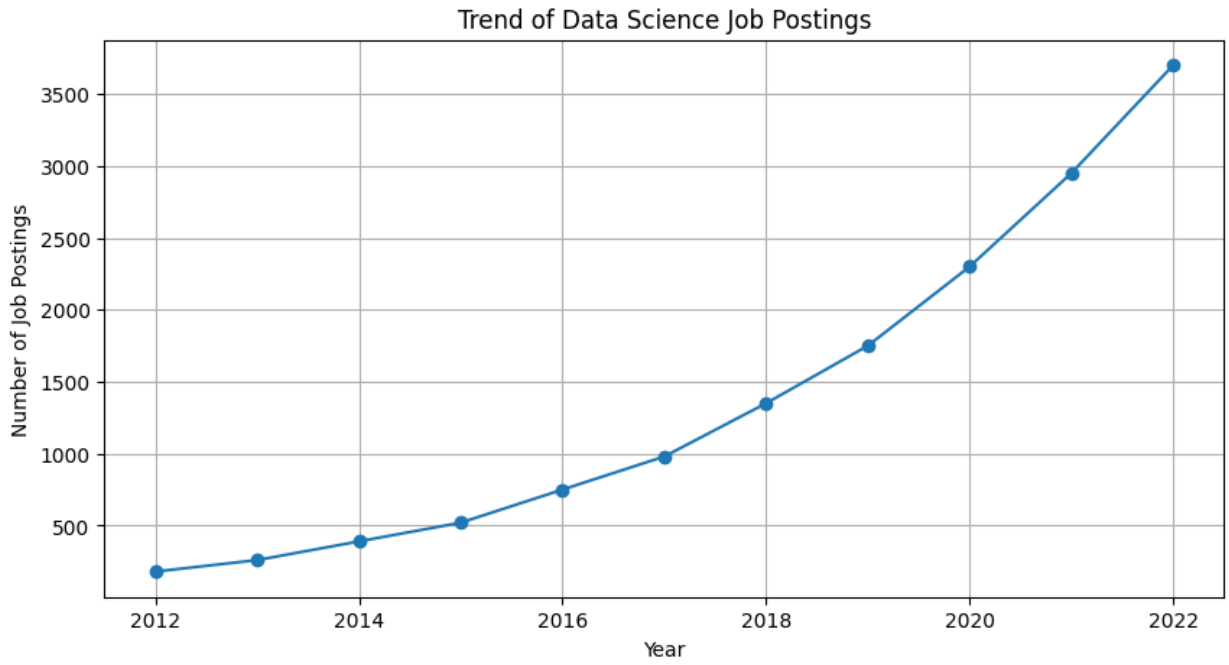
Aim:

To analyse the trend of data science job postings over a decade using Pandas and Matplotlib.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Year': list(range(2012, 2023)),
    'Job Postings': [180, 260, 390, 520, 750, 980, 1350, 1750, 2300, 2950, 3700]
}
df = pd.DataFrame(data)
print(df)
plt.figure(figsize=(10, 5))
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.grid(True)
plt.show()
```

OUTPUT:



RESULT:

The line graph shows an increasing trend on data science job postings from 2012 to 2022. The number of postings increased from 180 in 2012 to 3700 in 2022, indicating a significant growth in demand for data science professionals.

EXPERIMENT NO: 1.(b)

Aim:

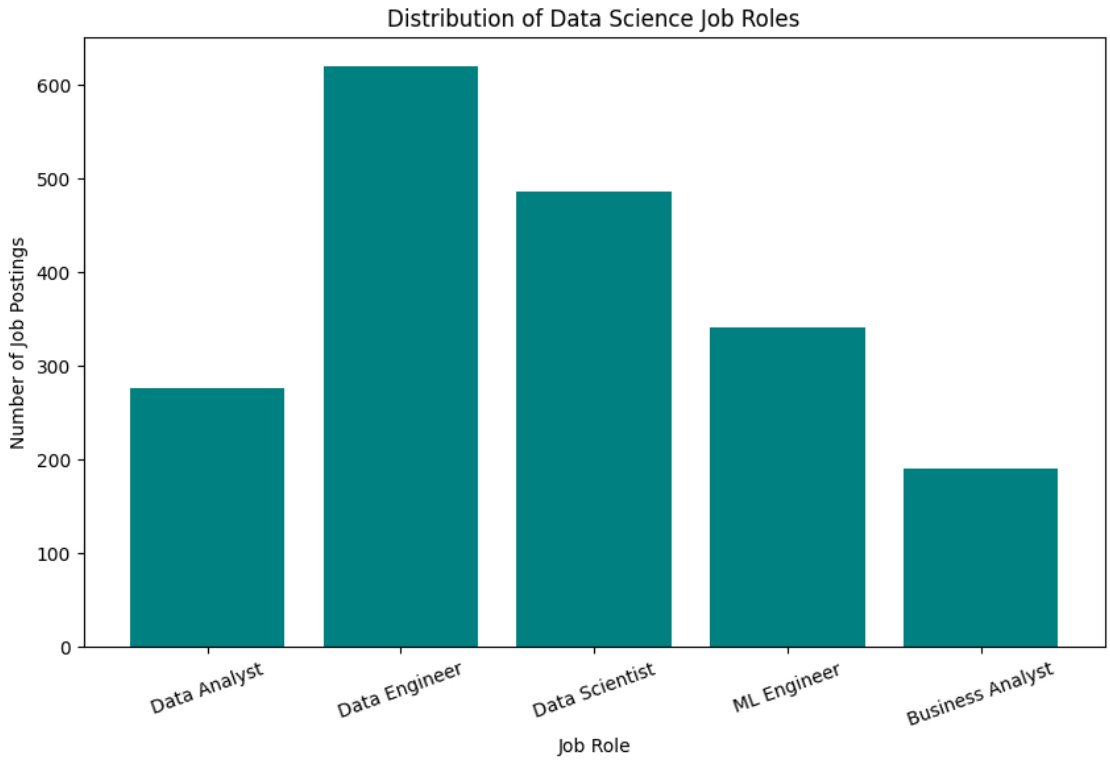
To analyse and visualize the distribution of different data science job roles using Pandas and Matplotlib.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    'Role': [
        'Data Analyst',
        'Data Engineer',
        'Data Scientist',
        'ML Engineer',
        'Business Analyst'
    ],
    'Count': [275, 620, 485, 340, 190]
}
df = pd.DataFrame(data)
print("Data Science Job Roles Dataset:")
print(df)
plt.figure(figsize=(10, 6))
plt.bar(
    df['Role'],
    df['Count'],
    color='teal')
```

```
plt.title('Distribution of Data Science Job Roles')
plt.xlabel('Job Role')
plt.ylabel('Number of Job Postings')
plt.xticks(rotation=20)
plt.show()
```

OUTPUT:



RESULT:

The bar graph shows that Data Engineer has the highest number of Job postings with 620. Business Analyst has the lowest with 190 postings. The graph helps compare the demand for different data-science-related roles.

EXPERIMENT NO: 1.(c)

Aim:

To differentiate between structured, unstructured and semi-structured data using suitable examples.

Code:

```
import pandas as pd
structured_data = pd.DataFrame({
    'ID': [101, 102, 103],
    'Name': ['David', 'Emma', 'John'],
    'Age': [28, 32, 26]
})
print("Structured Data:\n")
print(structured_data)
unstructured_data = (
    "Machine learning is a branch of artificial intelligence. "
    "It is used to analyze data and make predictions."
)
print("\nUnstructured Data:\n")
print(unstructured_data)
semi_structured_data = {
    'ID': 201,
    'Name': 'Sophia',
    'Details': {
        'Department': 'Data Science',
        'Experience': 3
```

```
    }
}
print("\nSemi-structured Data:\n")
print(semi_structured_data)
```

OUTPUT:

```
Structured Data:
   ID   Name  Age
0  101  David   28
1  102   Emma   32
2  103   John   26
Unstructured Data:
Machine learning is a branch of artificial intelligence. It is used to analyze data and make predictions.
Semi-structured Data:
{'ID': 201, 'Name': 'Sophia', 'Details': {'Department': 'Data Science', 'Experience': 3}}
```

RESULT:

The experiment successfully differentiated the three types of data based on their structure and format.

EXPERIMENT NO: 1.(d)

Aim:

To encrypt and decrypt sensitive data using the Fernet Cryptography library in Python.

Code:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()

cipher_suite = Fernet(key)

plain_text = b"Student ID: DS2024-105"

cipher_text = cipher_suite.encrypt(plain_text)

decrypted_text = cipher_suite.decrypt(cipher_text)

print("Original Data:", plain_text)

print("Encrypted Data:", cipher_text)

print("Decrypted Data:", decrypted_text)
```

OUTPUT:

Original Data: b'Student ID: DS2024-105'

Encrypted Data:
b'gAAAAABqhIyJpVUicEqCL5Ny5uEBwGJIGI3dI_p1oBt0HxlhdQU3GhY5IPhLH_uyzix_GdWCVw-e7wUIUOcY4POfrZtpWA-Y9N23HxEh8hDn152RaW29dV0='

Decrypted Data: b'Student ID: DS2024-105'

RESULT:

The sensitive data was successfully encrypted and decrypted using the Fernet cryptography library.